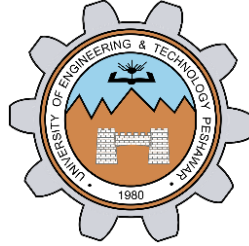


Static Web Design using HTML

LAB # 02



Spring 2024

CSE-403L Database Management Systems Lab

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Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Engr. Sumayyea Salahuddin

Date:

7th March 2024

Department of Computer Systems Engineering
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Objectives:

This lab aims at the understanding of:

- The basic syntax and structure of HTML.
- Identify the different types of headings and the appropriate usage for each.
- Understand how to create and use rules to style HTML elements.
- Learn how to use comments in HTML to provide additional information and context for the code.
- Understand how to structure paragraphs in HTML and use appropriate formatting tags.
- Understand how to use HTML styles to change the appearance of text and other elements.
- Learn how to create and format tables in HTML.
- Learn how to create and format lists in HTML.
- Understand how to create and use forms to collect user input.

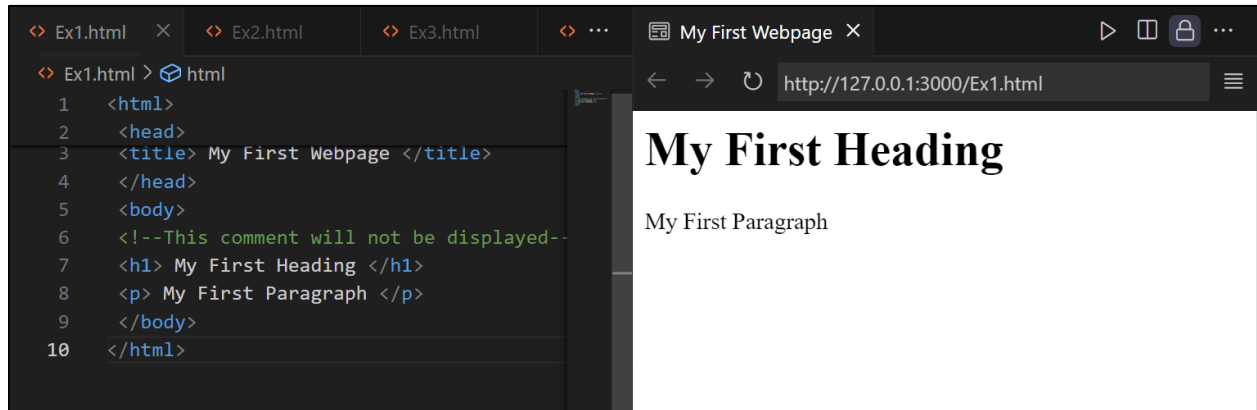
Task 2.1:

1. How HTML can be helpful for a SQL/NoSQL database? Write a short note on the paper.

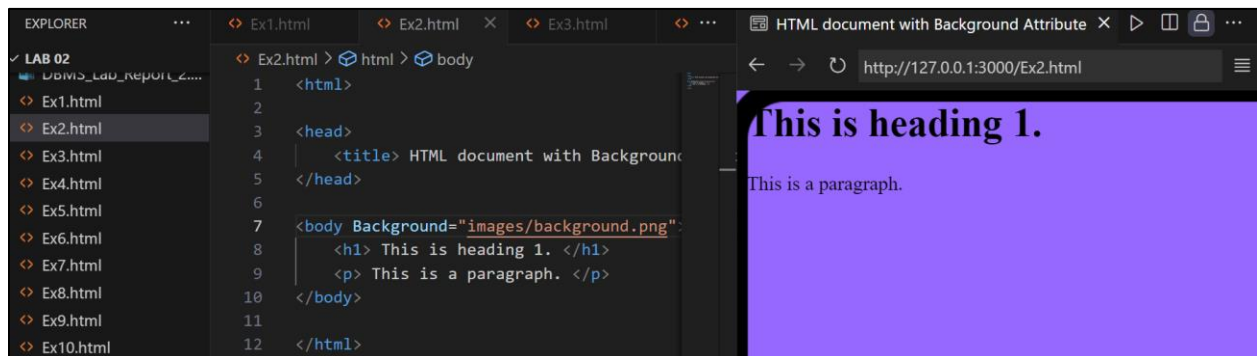
Ans: **HTML** serves as a crucial bridge between users and databases, whether you're dealing with **SQL** or **NoSQL** systems. In the context of **SQL databases**, HTML helps create user interfaces by enabling forms for data input. When users submit forms (such as login or registration), the server-side code processes the data and interacts with the SQL database. For **NoSQL databases**, HTML plays a role in building dynamic web pages that display data from unstructured or semi-structured sources. Whether it's rendering JSON data from a document-oriented NoSQL database or creating real-time interfaces, HTML ensures seamless interaction with databases.

2. Hands-On: Understand and go through all the examples provided in this lab. Try each one in your favorite IDE. Check output in browser and try to make some changes from your own.

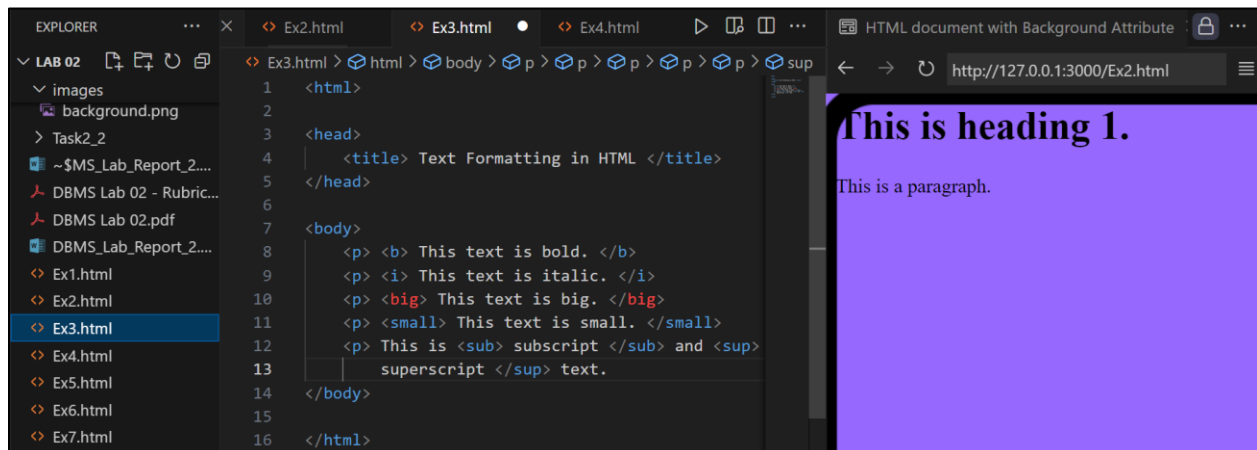
Example 1:



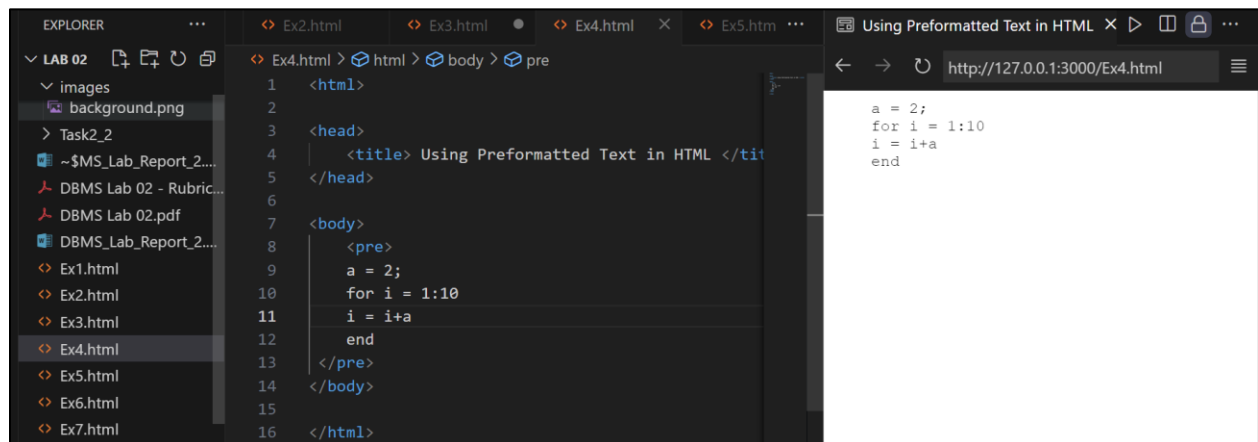
Example 2:



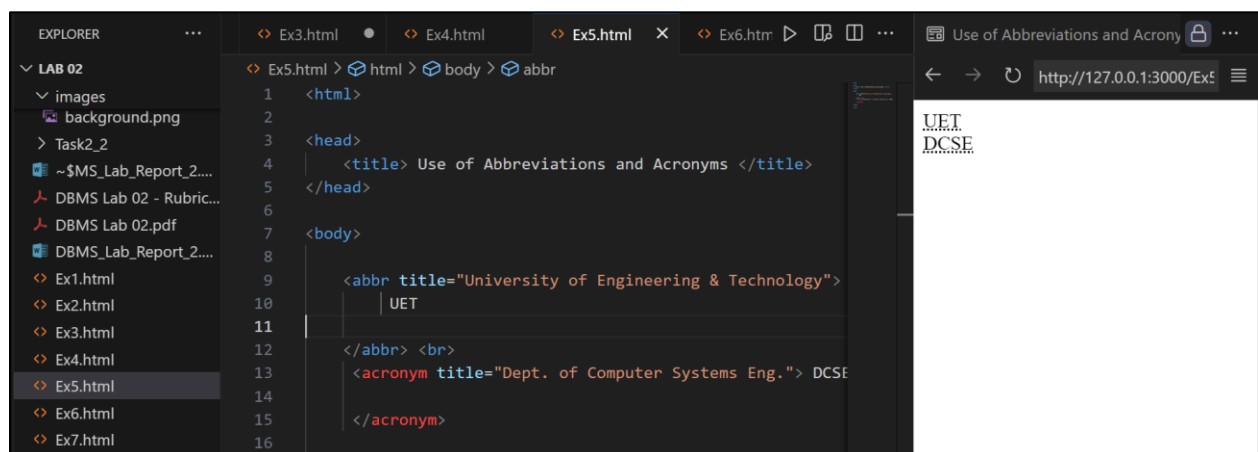
Example 3:



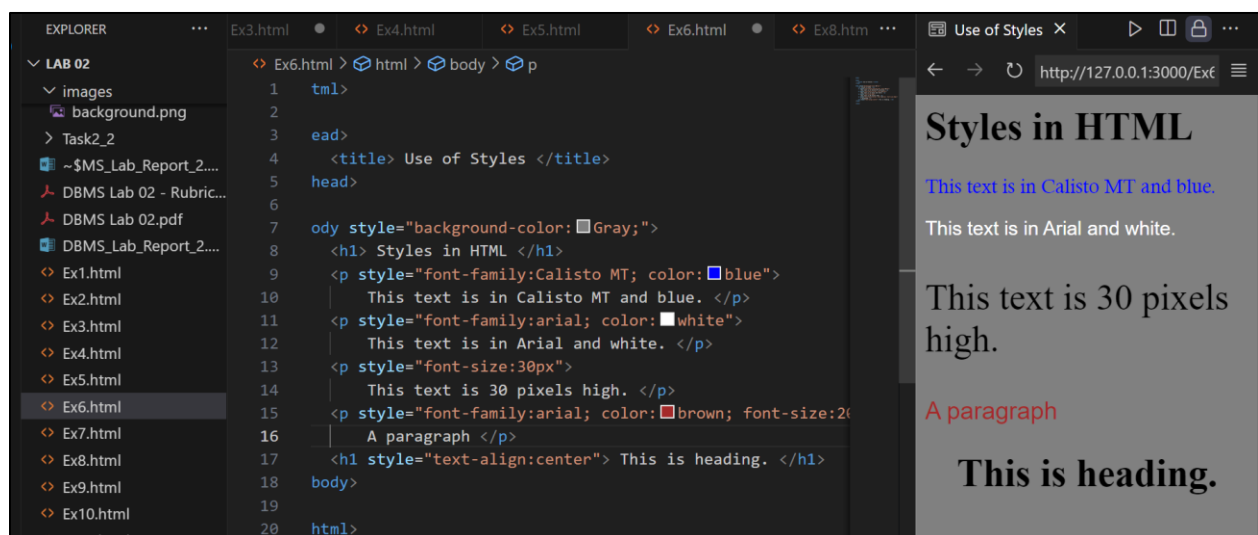
Example 4:



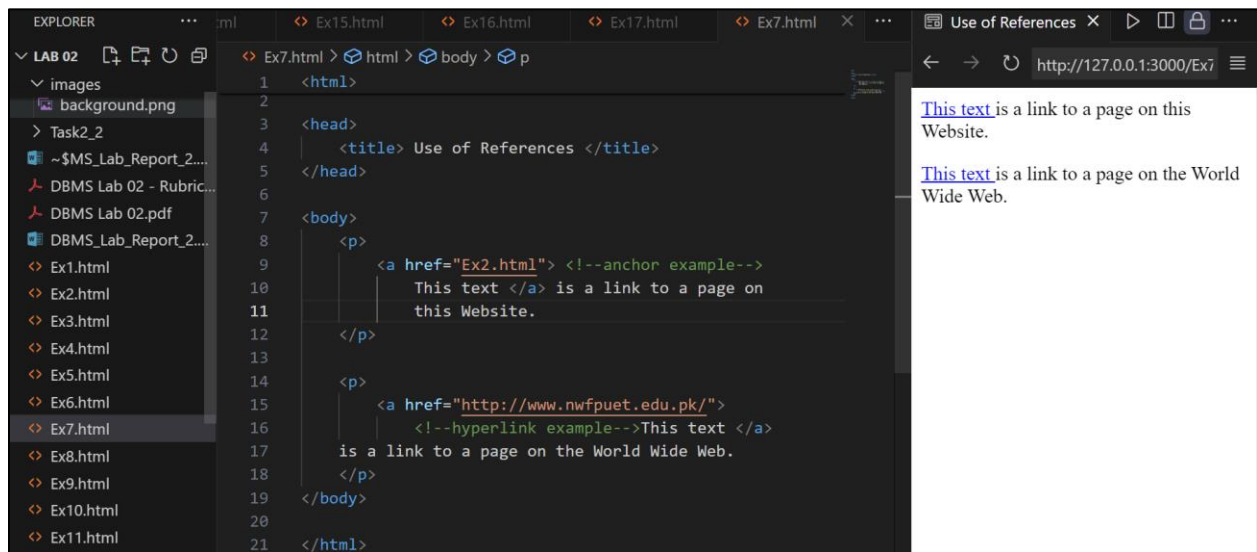
Example 5:



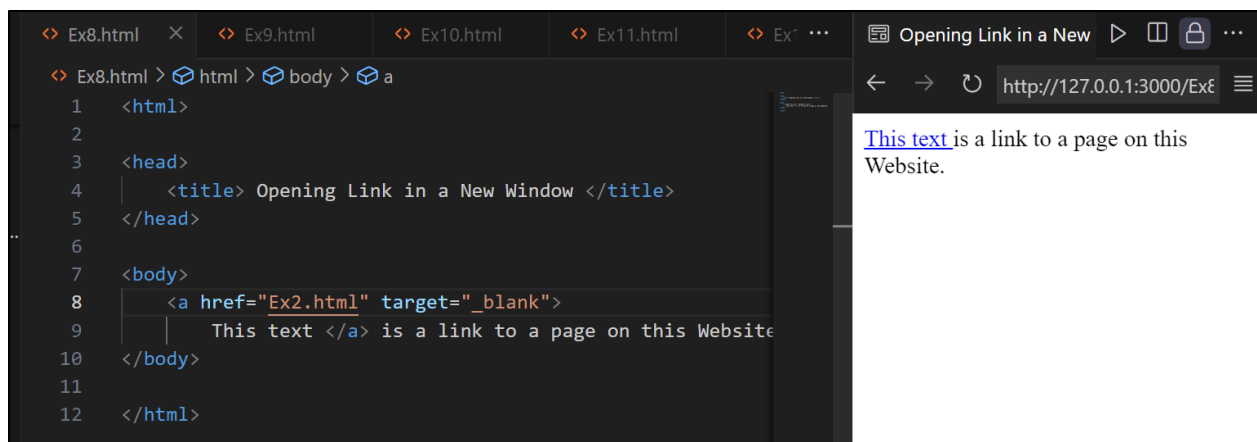
Example 6:



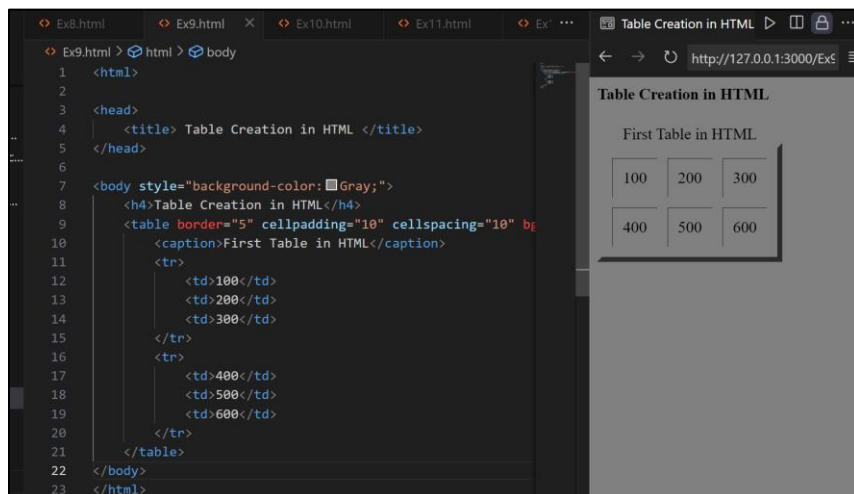
Example 7:



Example 8:



Example 9:



Example 10:

Ex8.htmlEx9.htmlEx10.htmlEx11.html

Ex10.html > html > body

```
1 <html>
2
3 <head>
4   <title> Table with Different Heading Styles </title>
5 </head>
6
7 <body style="background-color: Gray;">
8   <h4>Horizontal headers:</h4>
9   <table border="1">
10     <tr>
11       <th>Heading</th>
12       <th>Another Heading</th>
13     </tr>
14     <tr>
15       <td>row 1, cell 1</td>
16       <td>row 1, cell 2</td>
17     </tr>
18     <tr>
19       <td>row 2, cell 1</td>
20       <td>row 2, cell 2</td>
21     </tr>
22   </table>
23
```

Table with Different Heading Styl

http://127.0.0.1:3000/Ex1

Horizontal headers:

Heading	Another Heading
row 1, cell 1	row 1, cell 2
row 2, cell 1	row 2, cell 2

Vertical headers:

First Name:	Bill Gates
Telephone:	555 777 1854
Telephone:	555 777 1855
Telephone:	

```
23 <h4>Vertical headers:</h4>
24 <table border="1">
25   <tr>
26     <th>First Name:</th>
27     <td>Bill Gates</td>
28   </tr>
29   <tr>
30     <th>Telephone:</th>
31     <td>555 777 1854</td>
32   </tr>
33   <tr>
34     <th>Telephone:</th>
35     <td>555 777 1855</td>
36   </tr>
37   <tr>
38     <th>Telephone:</th>
39     <td>&nbsp;</td>
40   </tr>
41 </table>
42 </body>
43 </html>
```

Heading	Another Heading
row 1, cell 1	row 1, cell 2
row 2, cell 1	row 2, cell 2

Vertical headers:

First Name:	Bill Gates
Telephone:	555 777 1854
Telephone:	555 777 1855
Telephone:	

Example 11:

The screenshot shows a web browser with the address bar displaying `http://127.0.0.1:3000/Ex1`. The browser window is titled "Usage of Frames attribute". The main content area displays two examples of frames:

With frame="border":

First	Row
Second	Row

With frame="box":

First	Row
Second	Row

The background of the browser window is a dark-themed code editor showing the HTML code for Ex11.html. The code includes a `<body style="background-color: rgb(255, 255, 255);">` and two tables: one with `frame="border"` and one with `frame="box"`.

The screenshot shows a web browser with the address bar displaying `http://127.0.0.1:3000/Ex1`. The browser window is titled "Usage of Frames attribute". The main content area displays four examples of frames:

With frame="below":

First	Row
Second	Row

With frame="hsides":

First	Row
Second	Row

With frame="vsides":

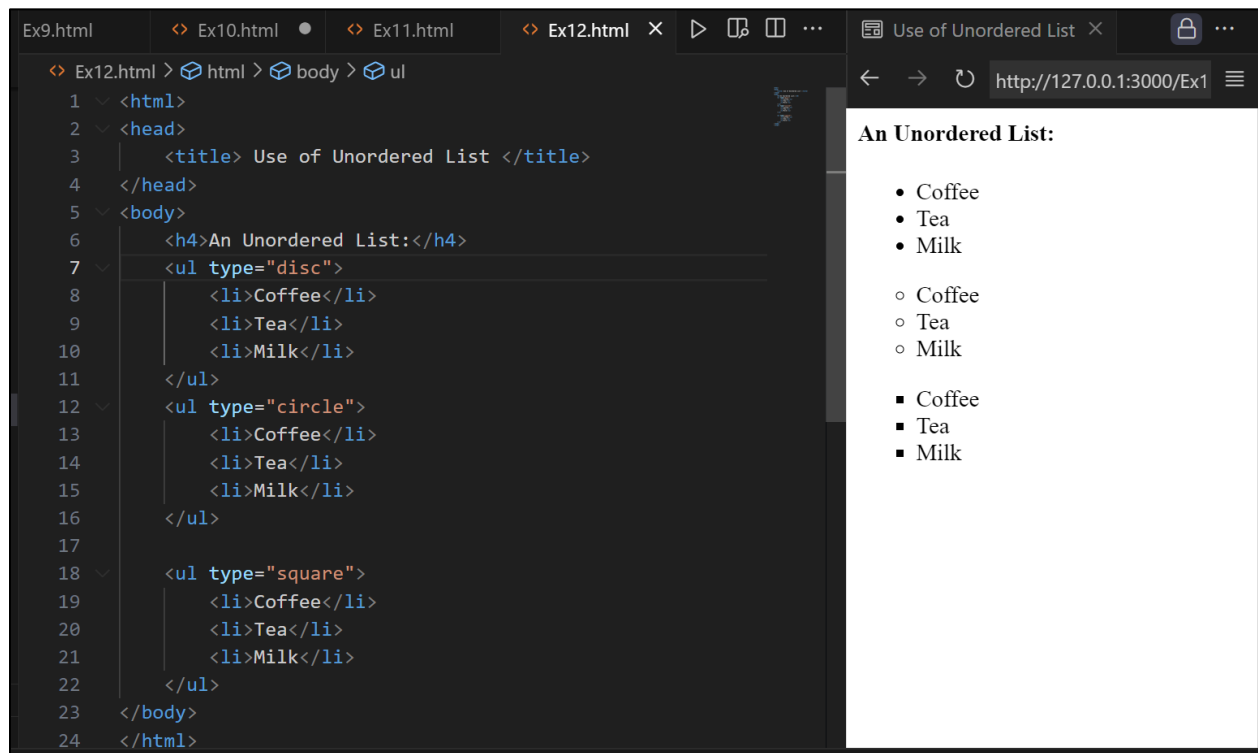
First	Row
Second	Row

With frame="lhs":

First	Row
Second	Row

The background of the browser window is a dark-themed code editor showing the HTML code for Ex11.html. The code includes a `<body style="background-color: rgb(255, 255, 255);">` and four tables: one with `frame="box"`, one with `frame="void"`, and two with `frame="border"`.

Example 12:



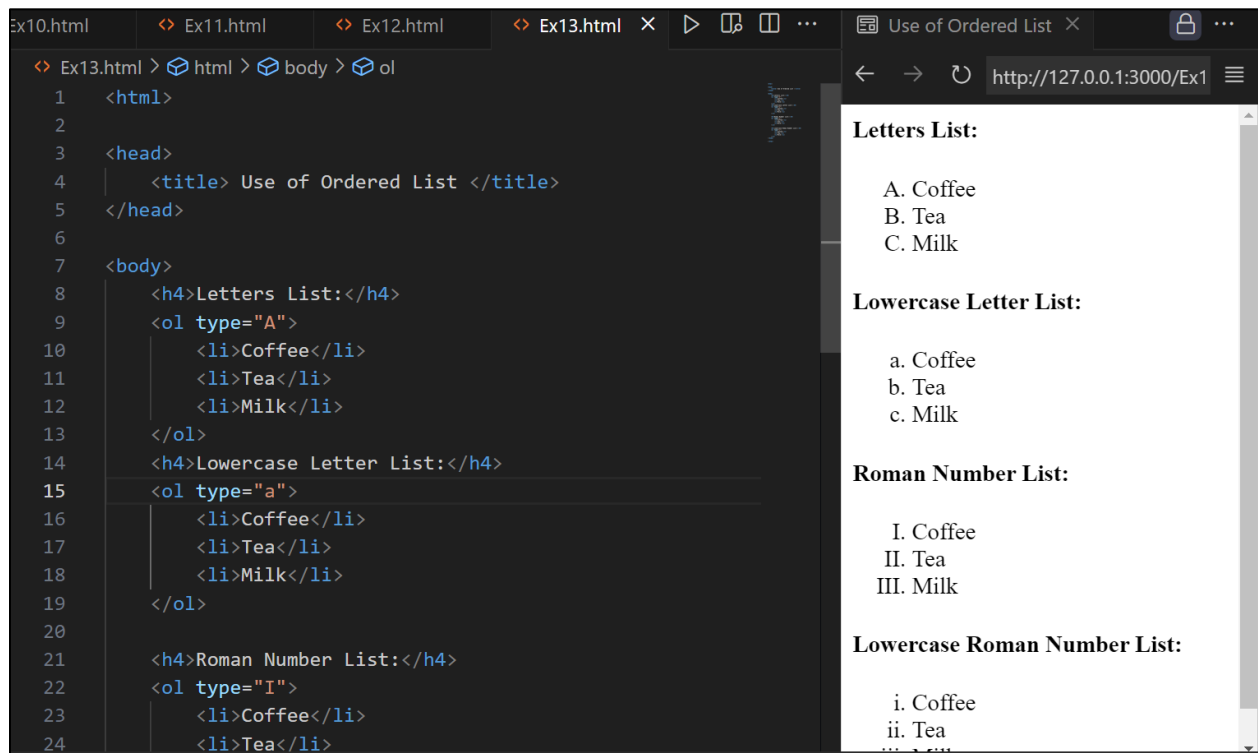
The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:3000/Ex1`. The browser tab is titled "Use of Unordered List". The main content area displays the heading "An Unordered List:" followed by three types of unordered lists:

- Disc markers:
 - Coffee
 - Tea
 - Milk
- Circle markers:
 - Coffee
 - Tea
 - Milk
- Square markers:
 - Coffee
 - Tea
 - Milk

The background shows a code editor with the following HTML code for `Ex12.html`:

```
<html>
<head>
  <title> Use of Unordered List </title>
</head>
<body>
  <h4>An Unordered List:</h4>
  <ul type="disc">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ul>
  <ul type="circle">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ul>
  <ul type="square">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ul>
</body>
</html>
```

Example 13:



The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:3000/Ex1`. The browser tab is titled "Use of Ordered List". The main content area displays the heading "Letters List:" followed by four types of ordered lists:

- Letters List:
 - A. Coffee
 - B. Tea
 - C. Milk
- Lowercase Letter List:
 - a. Coffee
 - b. Tea
 - c. Milk
- Roman Number List:
 - I. Coffee
 - II. Tea
 - III. Milk
- Lowercase Roman Number List:
 - i. Coffee
 - ii. Tea

The background shows a code editor with the following HTML code for `Ex13.html`:

```
<html>
<head>
  <title> Use of Ordered List </title>
</head>
<body>
  <h4>Letters List:</h4>
  <ol type="A">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ol>
  <h4>Lowercase Letter List:</h4>
  <ol type="a">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ol>
  <h4>Roman Number List:</h4>
  <ol type="I">
    <li>Coffee</li>
    <li>Tea</li>
  </ol>
</body>
```

```

21 <h4>Roman Number List:</h4>
22 <ol type="I">
23   <li>Coffee</li>
24   <li>Tea</li>
25   <li>Milk</li>
26 </ol>
27
28 <h4>Lowercase Roman Number List:</h4>
29 <ol type="i">
30   <li>Coffee</li>
31   <li>Tea</li>
32   <li>Milk</li>
33 </ol>
34 </body>
35
36 </html>

```

A. Coffee
B. Tea
C. Milk

Lowercase Letter List:

a. Coffee
b. Tea
c. Milk

Roman Number List:

I. Coffee
II. Tea
III. Milk

Example 14:

Ex14.html > html > body > dl > dt

```

1 <html>
2
3 <head>
4   <title> Use of Definition List </title>
5 </head>
6
7 <body>
8   <h4>A Definition List:</h4>
9   <dl>
10    <dt>Coffee</dt>
11    <dd>Black Hot Drink</dd>
12    <dt>Milk</dt>
13    <dd>White Cold Drink</dd>
14  </dl>
15 </body>
16
17 </html>

```

A Definition List:

Coffee
Black Hot Drink

Milk
White Cold Drink

Example 15:

```
1 <html>
2
3 <head>
4   <title> Form Creation in HTML </title>
5
6 </head>
7
8 <body>
9
10   <form action="">
11     First Name:
12     <input type="text" name="first name" />
13     <br />
14     Last Name:
15     <input type="text" name="last name" />
16     <br /> <br />
17
18     I've a bike:
19     <input type="checkbox" name="vehicle" value="bike"
20     I've a car:
21     <input type="checkbox" name="vehicle" value="car"
22     <br /> <br />
23
24     Male:
```

```
24   Male:
25   <input type="radio" name="sex" value="male" checked="" />
26   Female:
27   <input type="radio" name="sex" value="female" />
28   <br /> <br />
29
30   <input type="button" value="Hello World!" />
31   <br /> <br />
32
33   <select name="cars">
34     <option value="mehran">Mehran</option>
35     <option value="liana">Liana </option>
36     <option value="gimini">Gimini</option>
37   </select>
38   <br /> <br />
39
40 </form>
41 </body>
42
43 </html>
```

Example 16:

```
1 <html>
2
3 <head>
4   <title> Use of Fieldset, Legend, and Textarea Tags </title>
5 </head>
6
7 <body>
8   <fieldset>
9     <legend>text area legend</legend>
10    <textarea rows="10" cols="30">
11      The cat was playing in the garden.
12    </textarea>
13  </fieldset>
14 </body>
15
16 </html>
```

text area legend

The cat was playing in the garden.

Example 17:

```
1 <html>
2
3 <head>
4   <title> Use of Form Attributes </title>
5 </head>
6
7 <body>
8   <form name="input" action="html_form_submit.asp" method="post">
9     Username:
10    <input type="text" name="user" />
11    <input type="submit" value="Submit" />
12  </form>
13 </body>
14
15 </html>
```

Username:

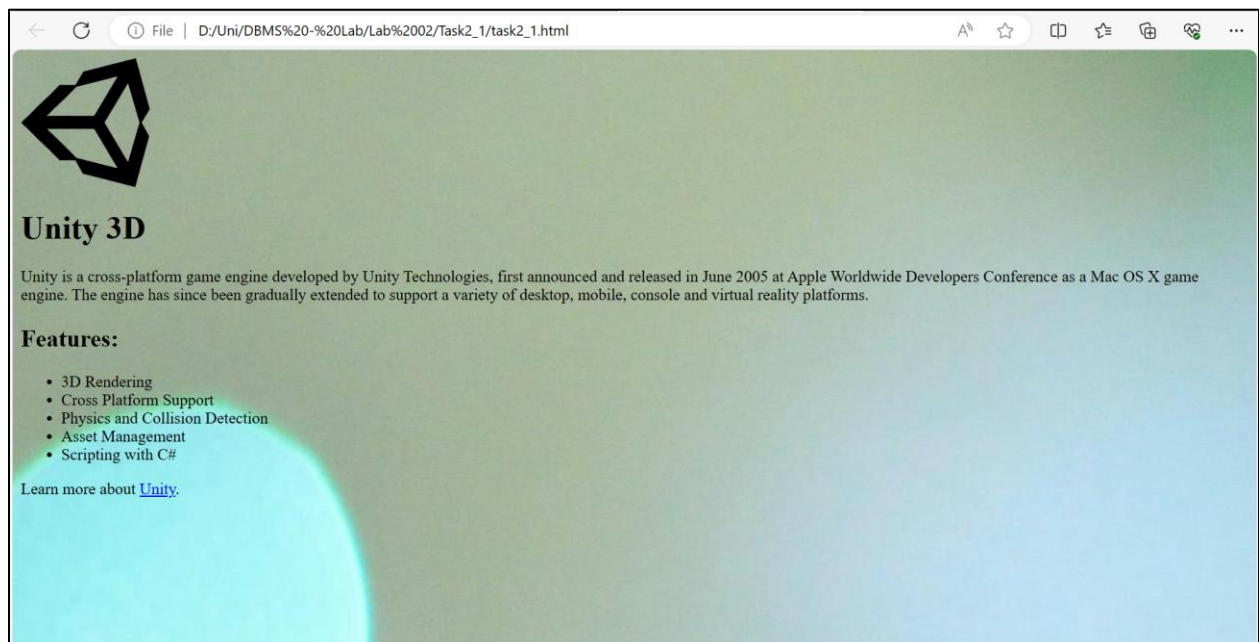
Submit

3. Hands-On: Create a basic web page for an application of your choice using HTML tags covered in this lab. Your web page can have a title, headings, paragraphs, lists, images, and links.

Code:

```
task2_1.html > task2_1.html X
task2_1.html > html > body > h2
1 <html>
2 <head>
3   <title>Unity 3D</title>
4 </head>
5 <body background="bg.jpeg">
6   
7   <h1>Unity 3D</h1>
8   <p>Unity is a cross-platform game engine developed by Unity Technologies, first announced in June 2005 at Apple Worldwide Developers Conference as a Mac OS X game engine. The engine has since been gradually extended to support a variety of desktop, mobile, console and virtual reality platforms.
9   <h2>Features:</h2>
10  <ul>
11    <li>3D Rendering</li>
12    <li>Cross Platform Support</li>
13    <li>Physics and Collision Detection</li>
14    <li>Asset Management</li>
15    <li>Scripting with C#</li>
16  </ul>
17  <p>Learn more about <a href="https://www.unity.com">Unity</a>.</p>
18 </body>
19 </html>
```

Output:



4. Evaluate yourself and write a note about your understanding of this lab.

Ans: Things Which I Learned and understood in this lab.

1. Basic Syntax and Structure of HTML:
2. Headings and Their Usage:
3. Styling HTML Elements:
4. Comments in HTML:
5. Paragraphs and Formatting:
6. HTML Styles:
7. Creating and Formatting Tables:
8. Creating and Formatting Lists:
9. Forms for User Input:

5. Make DBMS Lab Project group and share the name and registration number of group members on the paper.

Ans: Names of our group members are:

1. Shahzad Bangash(21PWCSE1980)
2. Suleman Shah(21PWCSE1983)
3. Ali Asghar(21PWCSE2059)

Task 2.2:

Code:

```
<> Task.html X <> Ex15.html ▶
<> Task.html > html
1  <html>|
2  <head>
3  |   <title> Sign Up/Login Form </title>
4  </head>
5
6  <body>
7  |   <!--This comment will not be displayed-->
8  |   <div align = "center">
9  |       <h1> Sign Up/Login Form </h1>
10 |
11 |       <form action="">
12 |
13 |           <input type="text" name="first name" placeholder="First Name" />
14 |           <br /> <br />
15 |
16 |           <input type="text" name="last name" placeholder="Last Name"/>
17 |           <br /> <br />
18 |
19 |           <input type="text" name="email" placeholder="Email"/>
20 |           <br /> <br />
21 |
22 |           <input type="password" name="password" placeholder="Enter new password"/>
23 |           <br /> <br />
24 |
25 |           <input type="password" name="password" placeholder="Enter new password"/>
26 |           <br /> <br />
27 |
28 |           <input type="confirm_password" name="confirm_password" placeholder="Confirm new pass" />
29 |           <br /> <br />
30 |
31 |           Male:
32 |           <input type="radio" name="sex" value="male" checked />
33 |           Female:
34 |           <input type="radio" name="sex" value="female" />
35 |           <br /> <br />
36 |
37 |           Do you want to recieve News from us?:
38 |           <input type="checkbox" name="news" value="bike" checked /> <br />
39 |           I accept the terms and conditions:
40 |           <input type="checkbox" name="vehicle" value="car" />
41 |           <br /> <br />
42 |           <br /><br />
43 |       </form>
44 |   </div>
45 </body>
46 </html>
```

Output:

The screenshot shows a web browser window with the address bar displaying 'D:/Uni/DBMS%20-%20Lab/Lab%202002/Task2_2/Task.html'. The page title is 'Sign Up/Login Form'. The form contains the following elements:

- First Name:
- Last Name:
- Email:
- Enter new password:
- Confirm new password:
- Gender: Male: ☒ Female: ☐
- Do you want to receive News from us?: ☒
- I accept the terms and conditions: ☐
- Sign Up:
- Sign In Instead:

Task 2.3:

Code:

```
< Task.html x Table.html
Task.html > html > body > table > tr > td > table
1 <html>
2 <head>
3 | <title>Tables</title>
4 </head>
5
6 <body>
7
8 <h1>Tables</h1>
9 Show<input type="text" placeholder=" "><br><br>
10 Search<input type="text" placeholder=" "><br><br>
11 <table border="1" width="400px">
12 | <tr>
13 | | <td width="100px">Ranks</td>
14 | | <td width="500px">Country</td>
15 | | <td width="100px">US $</td>
16 | | <td width="100px">Year</td>
17 | </tr>
18 | <tr>
19 | | <td>1</td>
20 | | <td>
21 | | <table>|
22 | | <tr>
23 | | <td></td>
```

Output:

File | D:/Uni/DBMS%20-%20Lab/Lab%2002/Task2_3/Task.html

Tables

Show

Search

Ranks	Country	US \$	Year
1	 Luxembourg	113.533	2011
1	 Qatar	98.329	2011
1	 Norway	97.255	2011
1	 Switzerland	81.181	2011
1	 United Arab Emirates	67.008	2011
1	 Australia	65.477	2011
1	 Denmark	59.928	2011
1	 Sweden	56.956	2011
1	 Canada	50.436	2011
1	 Netherlands	50.355	2011

Showing 1 to 10 of 186 records

Task 2.4:

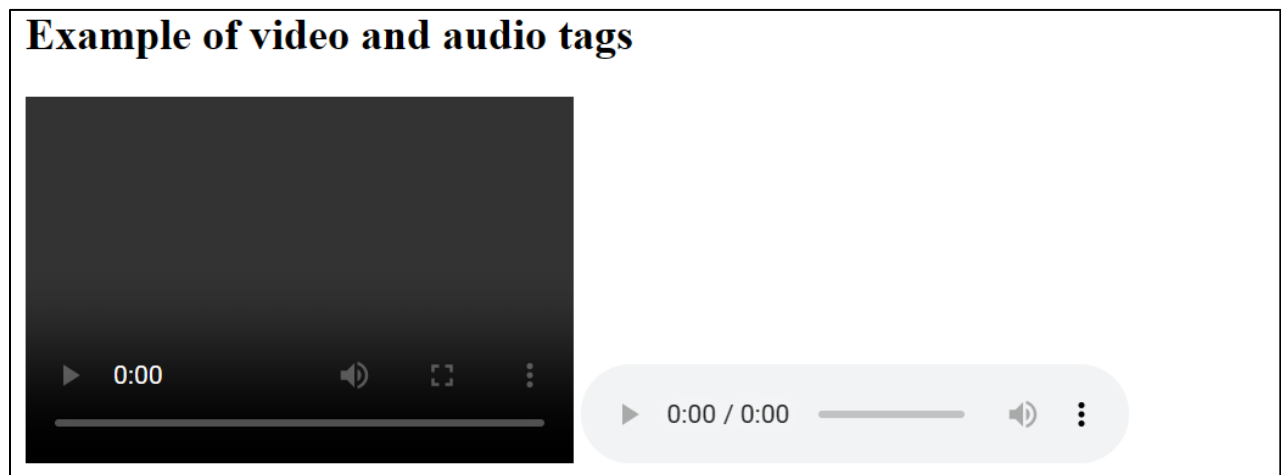
Audio and Video Tags:

HTML5 introduced the `<audio>` and `<video>` tags, allowing developers to embed audio and video content directly into web pages. We can use these tags to play audio files (such as music or podcasts) and video files (such as movies or tutorials).

Code:

```
E1.html x E2.html E3.html E4.html E5.html
E1.html > html > body > video
1 <!DOCTYPE html>
2 <html>
3
4 <body>
5   <h2>Example of video and audio tags</h2>
6   <video width="300" height="200" controls autoplay>
7     <source src="/html5/foo.ogg" type="video/ogg" />
8     <source src="/html5/foo.mp4" type="video/mp4" />
9     Your browser does not support the video element.
10  </video>
11  <audio controls autoplay>
12    <source src="/html5/audio.ogg" type="audio/ogg" />
13    <source src="/html5/audio.wav" type="audio/wav" />
14    Your browser does not support the audio element.
15  </audio>
16 </body>
17
18 </html>
```

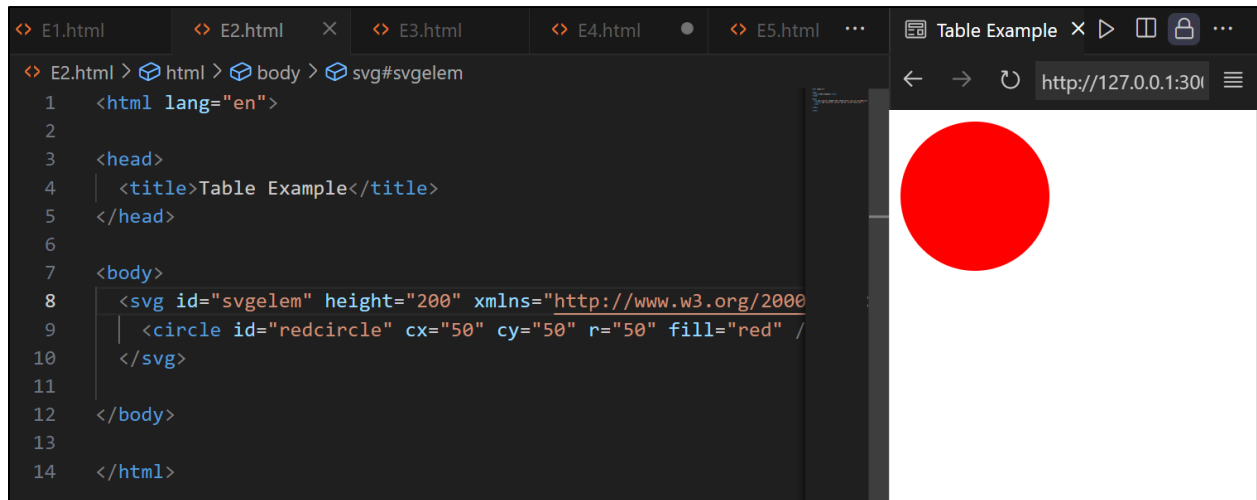
Output:



Vector Graphics (SVG):

Scalable Vector Graphics (SVG) is a powerful addition to HTML5. It allows us to create graphics with various shapes and colors using JavaScript. SVG graphics are scalable, easy to create, and editable. They also support interactivity and animation.

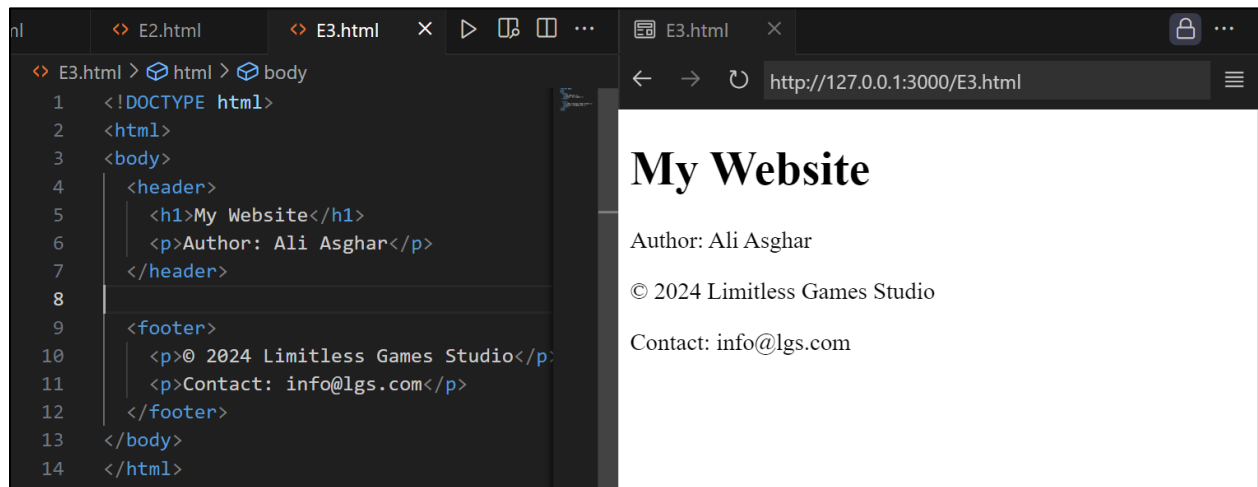
Code and output:



Header and Footer Elements:

HTML5 introduced semantic elements like `<header>` and `<footer>`. We can use `<header>` to define the header section of our web page (e.g., logo, authorship information), and `<footer>` for the footer section (e.g., copyright, contact information). These elements replace the need for `<div>` tags to identify header and footer sections.

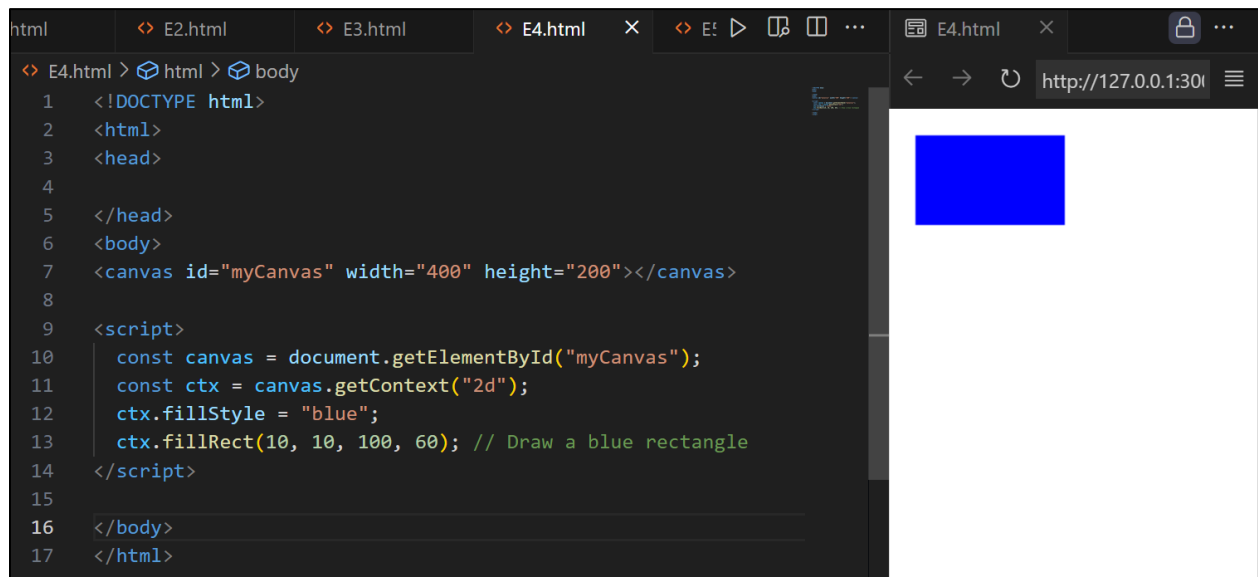
Code and output:

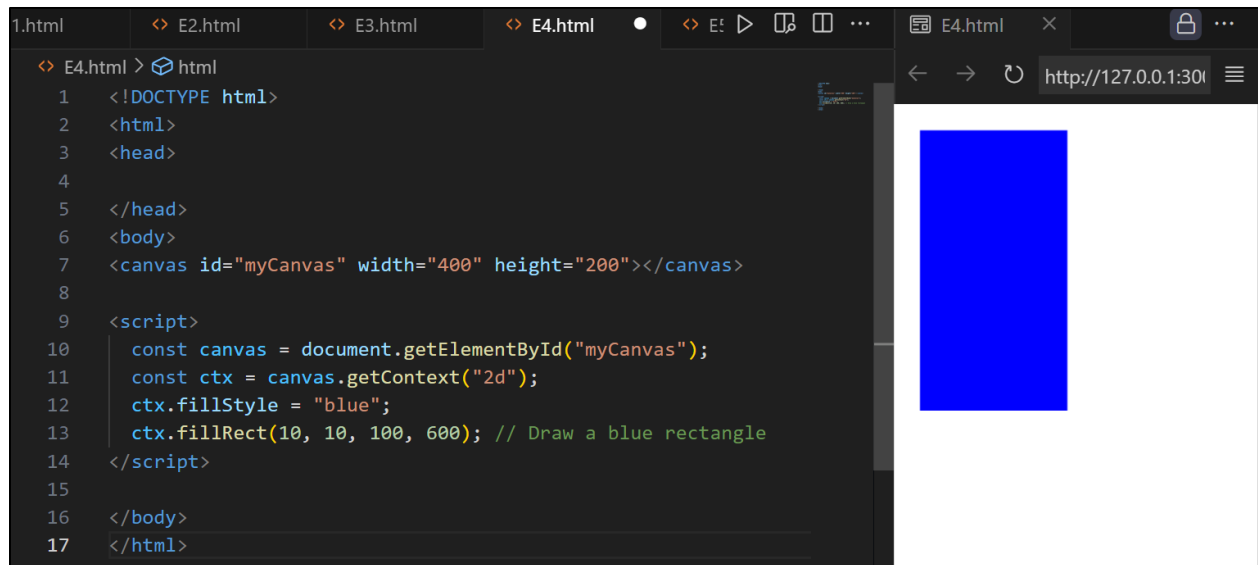


Canvas Element:

The `<canvas>` element allows dynamic rendering of graphics, animations, and interactive visualizations using JavaScript. We can draw shapes, images, and text directly on the canvas.

Code and output:



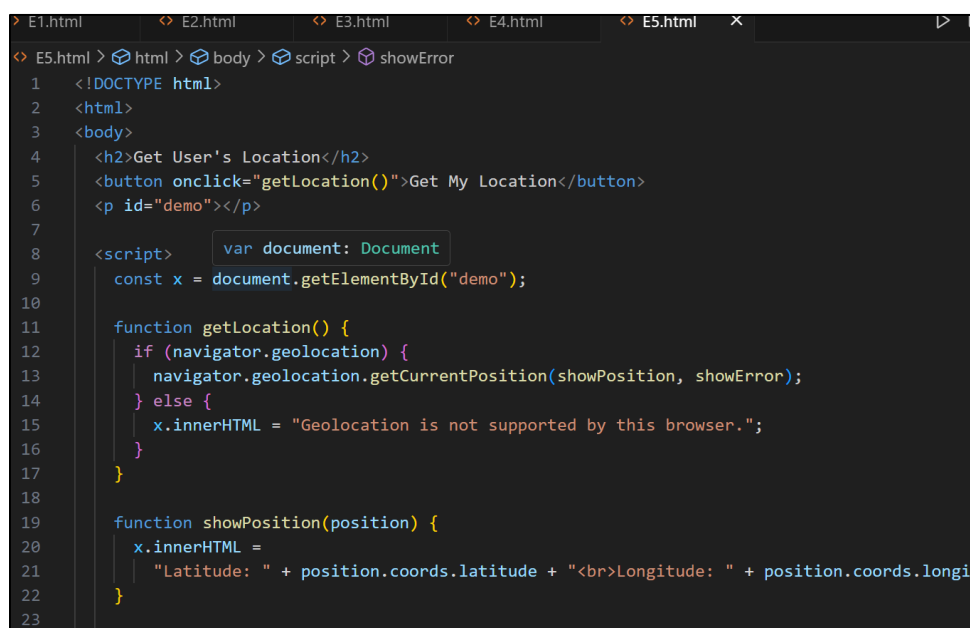


Geolocation feature:

HTML5 Geolocation feature allows us to determine a user's geographical position. This feature is particularly useful for creating location-aware web applications. Here's how it works:

- The HTML Geolocation API provides a way to retrieve the latitude and longitude coordinates of a user's current location.
- Keep in mind that this feature requires user approval, as it can potentially compromise privacy.
- Geolocation is most accurate on devices with GPS capabilities, such as smartphones.

Code:



```
23
24  ✓ function showError(error) {
25  ✓     switch (error.code) {
26  ✓         case error.PERMISSION_DENIED:
27             x.innerHTML = "User denied the request for Geolocation.";
28             break;
29  ✓         case error.POSITION_UNAVAILABLE:
30             x.innerHTML = "Location information is unavailable.";
31             break;
32  ✓         case error.TIMEOUT:
33             x.innerHTML = "The request to get user location timed out.";
34             break;
35  ✓         case error.UNKNOWN_ERROR:
36             x.innerHTML = "An unknown error occurred.";
37             break;
38         }
39     }
40 </script>
41 </body>
42 </html>
43
```

Output:

← ↻ ⓘ File | D:/Uni/DBMS%20-%20Lab/Lab%2002/Task2_4/E5.html

Get User's Location

Latitude: 30
Longitude: 70